METRIC

MIL-DTL-44411D
8 November 2012
SUPERSEDING
MIL-DTL-44411C
29 July 2005

DETAIL SPECIFICATION

INSECT REPELLENT, PERMETHRIN

This specification is approved for use by all departments and agencies of the Department of Defense.

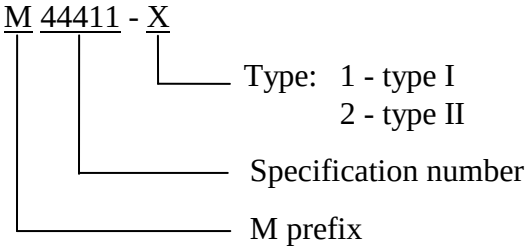
1. SCOPE

1.1 Scope. This specification covers permethrin insect/anthropod repellent for dilution in water and application to outer clothing, insect netting, and fabric.

1.2 Classification. The insect repellent covered by this specification should be of the following types, as specified (see 6.2).

Type I - In 151-milliliter (mL) bottles for dilution and hand spray application in the field.
Type II - In drums for industrial application to fabric.

1.3 Part or identifying number (PIN). The PIN to be used for insect repellent acquired to this specification is created as follows:



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AMSC N/A FSC 6840

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2. APPLICABLE DOCUMENTS

2.1 General. The documents listed in this section are specified in sections 3 and 4 of this specification. This section does not include documents cited in other sections of this specification or recommended for additional information or as examples. While every effort has been made to ensure the completeness of this list, document users are cautioned that they must meet all specified requirements of the documents cited in sections 3 and 4 of this specification, whether or not they are listed.

2.2 Government documents.

2.2.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract (see 6.2).

FEDERAL STANDARDS

FED-STD-595/24097	- Green, Semigloss
FED-STD-595/30051	- Brown, Flat or Lusterless
FED-STD-595/32169	- Orange, Flat or Lusterless

DEPARTMENT OF DEFENSE STANDARDS

MIL-STD-3010	- Test Procedures for Packaging Materials.

(Copies of these documents are available online at https://assist.dla.mil or from the Standardization Document Order Desk, 700 Robbins Avenue, Building 4D, Philadelphia, PA 19111-5094.)

2.2.2 Other government documents, drawings, and publications. The following other government documents, drawings, and publications form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract (see 6.2).

CODE OF FEDERAL REGULATIONS (CFR)

40 CFR 152	- Pesticide Registration and Classification Procedures.
49 CFR	- Transportation.

(Copies of these documents are available online at http://www.access.gpo.gov/ or from Superintendent of Documents, P.O. Box 371954, Pittsburgh, PA 15250-7954.)

2.3 Order of precedence. Unless otherwise noted herein or in the contract, in the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

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3. REQUIREMENTS

3.1 Insect repellent solution. The insect repellent concentrate for the type I and type II application methods shall meet the requirements specified in 3.1.1 through 3.1.3.

3.1.1 Composition. The insect repellent concentrate for the type I and type II application methods shall consist of: 40 to 42 percent by weight permethrin, U.S. Patent No. 4,024,163, (3-phenoxyphenyl)-methyl (±) cis/trans-3-(2,2-dichloroethenyl)-2,2-dimethyl-cyclopropanecarboxylate; cis/trans isomers mixture: min. 35 percent (±) cis and max. 65 percent (±) trans, and the balance 58 to 60 percent by weight inert petroleum distillates.

3.1.2 Appearance. The insect repellent concentrate shall be a homogeneous, translucent, yellow to amber colored solution free from insoluble matter when examined as specified in 4.5.1.

3.1.3 Emulsifiability. The insect repellent concentrate shall form a creamy emulsion and shall readily re-emulsify when tested as specified in 4.5.2.

3.2 Marking pen. The marking pen used to mark garments after treatment shall have sufficient ink to mark 12 sets (coats and trousers) of military battle dress uniforms. The marking pen ink shall be resistant to both dry cleaning and laundering.

3.3 Type I bottles and labels. Type I bottles and labels shall conform to the requirements in this section.

3.3.1 Type I plastic bottle, 151 mL. A quantity of 151 mL of insect repellent concentrate shall be supplied in a fluorinated, high density polyethylene (2F HDPE) bottle with ultraviolet (UV) blocker conforming to industrial packaging type 2 set forth in 49 CFR. The color of the bottle shall be Green, Semigloss (color number 24097), Brown, Flat or Lusterless (color number 30051), or Orange, Flat or Lusterless (color number 32169) in accordance with FED-STD-595.

3.3.1.1 Bottle screw cap. The screw cap shall be black or match the bottle color and shall be furnished with a resilient leak proof lining and facing that are highly resistant to degradation by the repellent concentrate and shall have a plastic seal to prevent inadvertent opening or loosening.

3.3.1.2 Leakage. The bottle and cap shall not leak when tested as specified in 4.4.4.

3.3.2 Labels and labeling information, type I. The labels shall be legibly and neatly printed, stenciled, lithographed, or silk-screened directly on the container if compatible with the type of material of which the container is made. As an alternate, the label information shall be neatly printed on paper stock in Green, Semigloss (color number 24097), Brown, Flat or Lusterless (color number 30051), or Orange, Flat or Lusterless (color number 32169) in accordance with FED-STD-595 and shall be sealed to the container with water-resistant adhesive. In all cases the ink shall be black, regardless of how the label information is applied to the container. The label shall not smear or be loosened from the container when tested as specified in 4.4.5.1 and 4.4.5.2. Labeling shall be in compliance with the requirements of the Federal Insecticide, Fungicide, and Rodenticide Act as amended and set forth in 40 CFR 152 (see

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figure 1). All directions and cautions on the labels shall be in accordance with and approved by the Pesticide Registration Division of the Environmental Protection Agency (EPA). In addition to the EPA approved label, a label containing the following information shall be affixed to each container:

a. Name, brand, or trademark under which the product is sold.
b. Name and address of the producer or registrant.
c. Product lot number.

3.4 Type II packaging and labeling. Specification of the packaging and labeling requirements for type II containers shall be the responsibility of the procuring and/or manufacturing organizations. This includes obtaining label approval from EPA.

3.6 Unit pack. Twelve 151-mL bottles of repellent and twelve marking pens (see 3.2) shall be unit packed unless otherwise specified in the acquisition document (see 6.2).

4. VERIFICATION

4.1 Classification of inspection. The inspection requirements specified herein are classified as a conformance inspection (see 4.3).

4.2 Inspection conditions. For the composition, appearance, and emulsifiability tests, a representative specimen of 100 mL shall be removed from each sample unit and placed in a clean, dry container labeled to identify the lot and container from which it was taken. Unless otherwise specified, all inspections shall be performed on a lot. The lot size shall be expressed in units of filled unit containers.

4.3 Conformance inspection. Conformance inspection shall be performed in accordance with inspection provisions set forth herein. The characteristics shown in 3.1 when tested in accordance with 4.4 shall constitute minimum inspections to be performed by the supplier prior to government acceptance or rejection by lot. Unless otherwise specified, sampling shall be conducted in accordance with table I. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of assuring that all products or supplies submitted to the government for acceptance comply with all requirements of the contract. Failure of any test by any sample shall be cause for rejection of the lot represented.

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TABLE I. Sampling.

Number in a lot	Number of samples
2 to 25	2
26 to 150	3
151 to 1,200	5
1,201 to 7,000	8
7,001 to 20,000	10
20,001 to 35,000	15
over 35,000	20

4.4 Test methods and procedures.

4.4.1 Component and material inspection. In accordance with 4.3, components and materials shall be inspected in accordance with all the requirements of referenced documents unless otherwise excluded, amended, modified, or qualified in this specification or applicable purchase document.

4.4.1.1 Material certification. A certificate of compliance may be acceptable as evidence that the permethrin meets the requirements of 3.1. When certificates of compliance are submitted, the government reserves the right to inspect such items to determine the validity of the certification.

4.4.1.2 Testing of Permethrin. If certificates of compliance are not supplied or not acceptable, the permethrin concentrate may be tested for conformance. One method is given in 6.4. Other methods are also acceptable.

4.4.2 End item visual examination. The unit packaged end items shall be examined for the defects defined in table II. Sampling shall be conducted in accordance with table I. Failure of any test by any sample shall be cause for rejection of the lot represented.

TABLE II. Visual examinations.

Examination	Defect
Markings (label)	Missing, incomplete, incorrect, illegible, not as specified.
Workmanship	Dirty, not uniform, contains foreign matter; any other defect affecting serviceability.
Unit pack	Not as specified herein or in the acquisition document.

4.4.3 Net contents examination. The end items shall be examined for net contents. Sampling shall be conducted in accordance with table I. The lot shall be acceptable if the average net content per container for all sample units examined is within ±2 percent of the amount specified on the label.

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4.4.4 Container leakage testing. Sampling shall be conducted in accordance with table I. The leakage test shall be conducted in accordance with the hot water technique of test method 5009 of MIL-STD-3010. A stream of air bubbles shall be considered evidence of leakage.

4.4.5 Labels.

4.4.5.1 Resistance to water, for type I. The unit container shall be completely immersed in water at a temperature between 20 °C and 27 °C for 1 hour. Remove the unit container and rub fingers over the face of the label to determine if the label either smears or loosens to the point that it can be moved. Any smearing or loosening of the label constitutes a failure.

4.4.5.2 Resistance to insect repellent, for type I. The unit container shall be completely immersed in the type and concentration of repellent contained therein at a temperature between 20 °C and 27 °C for 1 hour. Remove the unit container and rub fingers over the face of the label to determine if the label either smears or loosens to the point that it can be moved. Any smearing or loosening of the label constitutes a failure.

4.5 Methods of inspection.

4.5.1 Appearance. Examine the insect repellent for color homogeneity and the presence of suspended particles or sediment.

4.5.2 Emulsifiability. Add 5 mL of insect repellent to a 100-mL graduated cylinder containing 85 mL of distilled water. Agitate gently by inverting the cylinder 10 times. A creamy emulsion shall be formed. Let stand for 15 minutes. If any separation has occurred, re-invert the cylinder gently 10 times, let stand for 15 minutes, and examine for homogeneity of the emulsion. The test shall be done at room temperature.

5. PACKAGING

5.1 Packaging. For acquisition purposes, the packaging requirements shall be as specified in the contract or order (see 6.2). When packaging of materiel is to be performed by DoD or in-house contractor personnel, these personnel need to contact the responsible packaging activity to ascertain packaging requirements. Packaging requirements are maintained by the inventory control point's packaging activities within the military service or defense agency, or within the military service's system commands. Packaging data retrieval is available from the managing military department's or defense agency's automated packaging files, CD-ROM products, or by contacting the responsible packaging activity.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

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6.1 Intended use. The type I insect repellent is intended for use in treating individual outer clothing units, such as the battle dress uniform, as well as insect netting and tents. The type II insect repellent is intended for use in industrial impregnation of fabric used to manufacture the battle dress uniform and is not intended for direct purchase by the government.

6.2 Acquisition requirements. Acquisition documents should specify the following:

a. Title, number, and date of this specification.
b. Type required (see 1.2).
c. The specific issue of individual documents referenced (see 2.2.1 and 2.2.2).
d. Packaging requirements (see 3.6 and 5).

6.3 Labels. An example of a label approved by the EPA for type I insect repellent is provided as figure 1.

6.4 Gas chromatographic method of analysis for determination of permethrin.

6.4.1 Scope. This procedure describes a gas chromatographic method for determining permethrin. The active ingredients are determined by injecting a solution of the analytes and acetone into the chromatograph. Separation is obtained and peak areas are compared to an external standard containing a known amount of the analyte.

6.4.2 Reagents/gases.

a. AgrEvo Environmental Health, Permanone® 40 Manufacturer's Concentrate, standard.
b. Acetone, HPLC pure.
c. High purity helium.
d. Compressed air.
e. Hydrogen.

6.4.3 Apparatus.

a. Gas chromatograph: Hewlett Packard Model 5890 or equivalent for data manipulation, gas chromatograph equipped with flame ionization detector (FID) and autosampler.
b. Column: A 6-foot by 2-mm inside diameter glass column.
c. Packing: A 3 percent OV-101 on 100/120 Chromosorb W-HP.
d. Hewlett Packard Chemstation.

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6.4.4 Conditions.

a. Flow (He): 30 mL/min.
b. Flow (H_2): 30 mL/min.
c. Flow (Air): 400 mL/min.
d. Initial temperature: 230 °C.
e. Initial time: 0.0 min.
f. Temperature program rate: 10 degrees/min.
g. Final temperature: 245 °C.
h. Final time: 7.0 min.
i. Run time: 8.5 min.
j. Injector temperature: 250 °C.
k. Detector temperature: 275 °C.
l. Injection volume: 2 µl.

6.4.5 External integrator settings.

a. Attenuation: 7.
b. Threshold: 5.
c. Peak width: 0.04.
d. Chart speed: 0.50 cm/min.

6.4.6 Standard preparation. In a 50-mL volumetric flask, weigh 50 mg of AgrEvo Environmental Health, Permanone® 40 Manufacturing Concentrate to the nearest 0.001 gram and dilute to volume with acetone. Transfer a small portion to a sample vial and label with contents and date. Place the stopper back on the volumetric flask and label it accordingly.

6.4.7 Sample preparation. In a 50-mL volumetric flask, weigh 125 mg of insect repellent sample to the nearest 0.001 gram and dilute to volume with acetone. Transfer a small portion to a sample vial and label with contents and date. Place the stopper back on the volumetric flask and label it accordingly.

6.4.8 Procedure. After the chromatograph is stable, make three or more injections of the standard and average the area counts. The injections should be repeatable to within 1 percent. Do the same for the sample. Calculate the percent purity by using the following formula:

$$\text{Percent purity} = \frac{\text{average sample area X weight of standard X 100}}{\text{average standard area X weight of sample}}$$

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6.5 Subject term (key word) listing.

arthropod
battle dress uniform
chiggers
clothing
emulsion
insecticide
lice
mosquitoes
netting
pesticide
petroleum distillate
pyrethroid
sand flies
tent
ticks

6.6 Changes from previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extent of the changes.

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INSECT/ARTHROPOD REPELLENT FABRIC TREATMENT

FOR USE BY CERTIFIED OR TRAINED PERSONNEL ONLY
FOR MILITARY USE ONLY
OUTER CLOTHING AND CONSTRUCTED FABRIC ITEMS ONLY

ACTIVE INGREDIENT:

Permethrin* ……	40.00%
OTHER INGREDIENTS** ……	60.00%
TOTAL	100.00%

* (3-phenoxyphenyl) methyl (±) cis/trans-3-(2.2-dichloroethenyl)-2.2-dimethyl-cyclopropanecarboxylate; cis/trans isomers ratio: min. 35% (±) cis and max. 65% (±) trans
** Contains petroleum distillates

KEEP OUT OF REACH OF CHILDREN
DANGER
SEE BACK PANEL FOR ADDITIONAL PRECAUTIONS

EPA Reg. No. 63120-1
EPA Est. No.

Produced and Packaged for:
DEPARTMENT OF DEFENSE
DEFENSE SUPPLY CENTER RICHMOND
RICHMOND, VIRGINIA 23297-5000

NET CONTENTS: 5.1 fl. oz. (151 ml)
Batch:

FIGURE 1. Approved EPA label for type I bottles.

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PRECAUTIONARY STATEMENTS
HAZARDS TO HUMANS AND DOMESTIC ANIMALS

DANGER

Corrosive. Causes irreversible eye damage. Do not get in eyes or on clothing. Wear protective eyewear (goggles, face shield, or safety glasses). Harmful if swallowed. Wash thoroughly with soap and water after handling fluid, container, or wet treated clothing and before eating and smoking. Remove contaminated clothing and wash clothing before use.

FIRST AID

IF IN EYES:	• Hold eye open and rinse slowly and gently with water for 15-20 minutes. • Remove contact lenses, if present, after the first 5 minutes, then continue rinsing eyes. • Call a poison control center or doctor for treatment advice.
IF SWALLOWED:	• Immediately call a poison control center or doctor for treatment advice. • Do not induce vomiting unless told to do so by a poison control center or doctor. • Do not give any liquid to the person. • Do not give anything by mouth to an unconscious person.
Have the product container or label with you when calling a poison control center or doctor or when going for treatment. You may also call the National Pesticide Information Center at 1-800-858-7378 for emergency medical treatment advice.	
NOTE TO PHYSICIAN: This product contains a pyrethroid and petroleum distillates. Vomiting may cause aspiration pneumonia. This product can cause severe eye damage. Probable mucosal damage may contraindicate the use of gastric lavage.	

ENVIRONMENTAL HAZARDS

This pesticide is extremely toxic to fish and other aquatic organisms. Keep out of lakes, ponds, or streams. Do not contaminate water by cleaning of equipment or disposal of wastes.

PHYSICAL OR CHEMICAL HAZARDS

Combustible. Do not use or store near heat or open flame.

DIRECTIONS FOR USE

It is a violation of Federal law to use this product in a manner inconsistent with its labeling.

This product may only be used by certified or trained military personnel to treat outer clothing (field uniform), and constructed fabric items (such as netting, non-vinyl tents and shelters, ground cloths, and vehicle covers). Treated clothing and constructed fabric items provide protection to covered areas of the body from bites of mosquitoes, ticks, chiggers, human body lice, and sand flies.

Do not apply to skin. Keep away from food, eating utensils and water supplies. Thoroughly wash any items contaminated with this product. Do not use on waterproof fabrics.

DO NOT APPLY TO UNDERWAER, T-SHIRT, OR CAP. Do not retreat uniform; one treatment is effective for the combat life of the field uniform (approximately 50 launderings).

Spray on fabric for a treatment level of 0.52% weight by weight of active ingredient.

NSN 6840-01-334-2666 Military Specification MIL-DTL-44411

FIGURE 1. Approved EPA label for type I bottles - Continued.

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Procedure: Wear protective eyewear, gloves, and respirator when mixing, loading and spraying. Empty entire contents of permethrin container into 2 gallons of clean water. Agitate and apply, using a sprayer equipped with a fan or cone nozzle and pressurized to produce a consistent discharge rate and even coverage. FOR FIELD UNIFORMS: Spread uniforms on the ground and evenly spray both sides of each uniform. Entire mixture must be used to properly treat 8 field uniforms. Sprayer must be calibrated to deliver 32 fl. oz. of mixture to each uniform (pants and blouse or coveralls). Allow uniform to dry completely, then mark inside jacket and trousers with date of treatment. FOR BED NETS: Spread netting on the ground. Spray entire outer surface of net to obtain even coverage. FOR TENTAGE: Spray entryways and interior surfaces (walls, ceiling, floor) of non-vinyl tent to obtain even coverage without runoff.

STORAGE AND DISPOSAL

Do not contaminate water, food, or feed by storage and disposal. **Pesticide Storage:** Do not store at temperature below 40°F (4.5°C). Store product in original container in a locked storage area. Store and transport in upright position. **Pesticide Disposal: If empty:** Do not reuse this container. Place in trash. **If partially filled:** Call your local solid waste agency or 1-800-CLEANUP for disposal instructions. Never pour unused product down any indoor or outdoor or on the ground.

FIGURE 1. Approved EPA label for type I bottles - Continued.

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Army – GL
Air Force – 68
DLA – GS

Preparing Activity:
DLA – GS3

(Project 6840-2012-002)

Review Activities:
Air Force – 11
Army – MD1
AFPMB – PM

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